Homework 21: Determinants and Elementary Operations

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BCS A, E, F, G

Theorem 1. ***Elementary Row Operations and Determinants.***

Let A and B be square matrices.

1. When B is obtained from A by interchanging two rows of A, $\det(B) = -\det(A)$.

2. When B is obtained from A by adding a multiple of a row of A to another row of A, $\det(B) = \det(A)$.

3. When B is obtained from A by multiplying a row of A by a nonzero constant c, $\det(B) = c\det(A)$.

Matrices and Zero Determinants.

Theorem 2. ***Conditions That Yield a Zero Determinant.***

If A is a square matrix and any one of the conditions below is true, then $\det(A) = 0$.

1. An entire row (or an entire column) consists of zeros.

2. Two rows (or columns) are equal.

3. One row (or column) is a multiple of another row (or column).

Problem 1. ***Finding a Determinant Using Elementary Row Operations.***

Find all determinant of A using elementary *row* operations.

$$A = \begin{bmatrix} 0 & -7 & 14 \\ 1 & 2 & -2 \\ 0 & 3 & -8 \end{bmatrix}.$$

In evaluating a determinant, it is occasionally convenient to use elementary column operations.

Problem 2. ***Finding a Determinant Using Elementary Column Operations.***

Find all determinant of A using elementary *column* operations.

$$A = \begin{bmatrix} -1 & 2 & 2 \\ 3 & -6 & 4 \\ 5 & -10 & -3 \end{bmatrix}.$$

Problem 3. ***A Matrix with a Zero Determinant.***

Find all determinant of A using *"Theorem 2"*.

$$A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & -1 & 0 \\ 0 & 18 & 4 \end{bmatrix}.$$

Problem 4. ***Finding a Determinant.***

Use any convenient method to find the determinant of A.

$$A = \begin{bmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & -3 \\ 1 & 1 & 3 & 2 & 0 \end{bmatrix}.$$

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Good Luck